Medical Insurance Charges Prediction with MLPs

Six fully connected networks (PyTorch) are trained to predict log(charges + 1) from age, sex, BMI (Box-Cox transformed), number of children, smoking status and region on the Medical Cost Personal dataset (1,338 records, 80/20 train/test split, random_state=42).

All figures, tables and numbers in this report are taken directly from the outputs of the executed notebook notebooks/insurance_charges_mlp.ipynb.

Execution: complete end-to-end run of the notebook on CPU (Python 3.11, seed 42, 2,000 full-batch epochs per model). This is not a partial run.

Final-epoch metrics on the log-charges target.

model	train_mse	test_mse	train_r2	test_r2
SimpleMLP1	0.3625	0.3428	0.5633	0.6187
SimpleMLP2	0.1789	0.1601	0.7844	0.8219
SimpleMLP3	0.1652	0.1457	0.8010	0.8380
SimpleMLP4	0.1378	0.1297	0.8340	0.8557
SimpleMLP5	0.1262	0.1271	0.8479	0.8586
EnhancedMLP	0.1656	0.1430	0.8004	0.8410

EnhancedMLP test-set metrics after mapping predictions back to dollars.

```
  MAE_usd: 3,078.4762
 RMSE_usd: 6,209.9162
   R2_usd: 0.7516
   R2_log: 0.8410
```

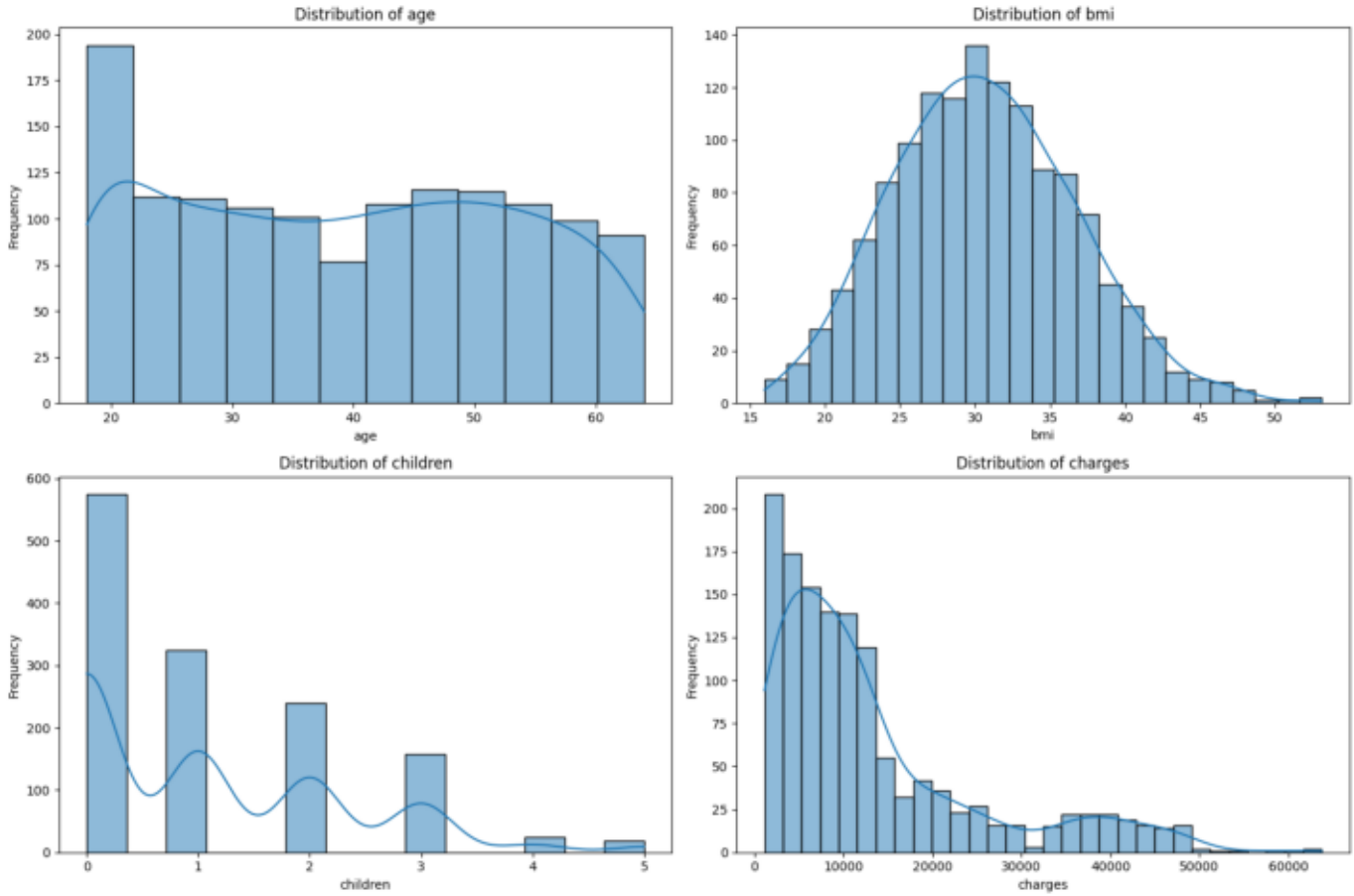


Figure 1. Distributions of the continuous features.

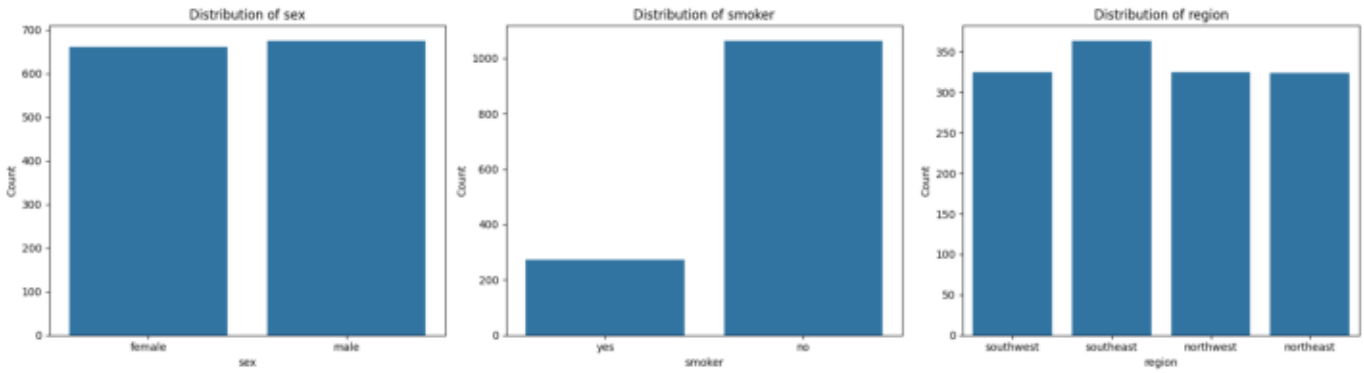


Figure 2. Distributions of the categorical features.

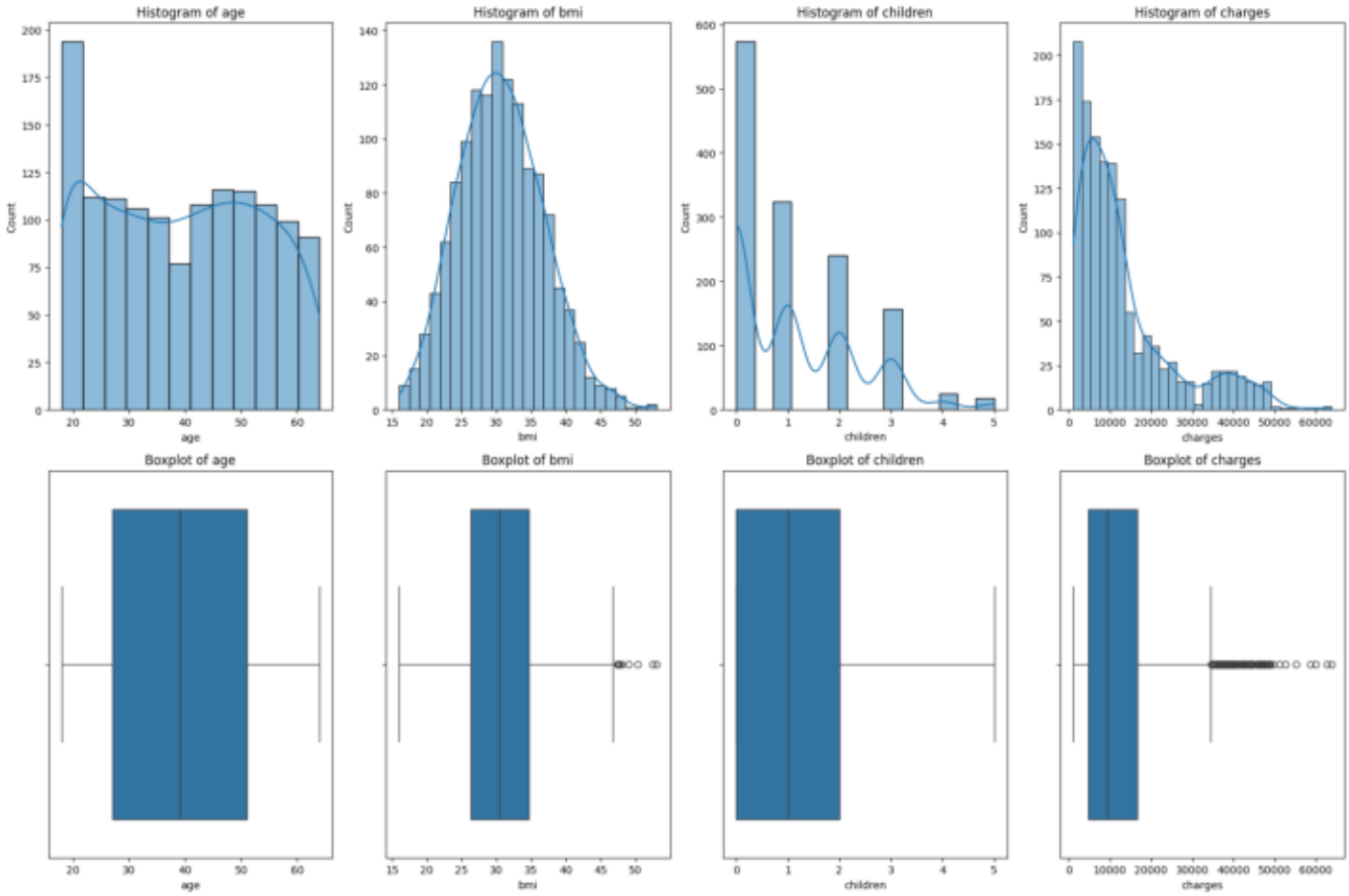


Figure 3. Histograms and boxplots of the continuous features.

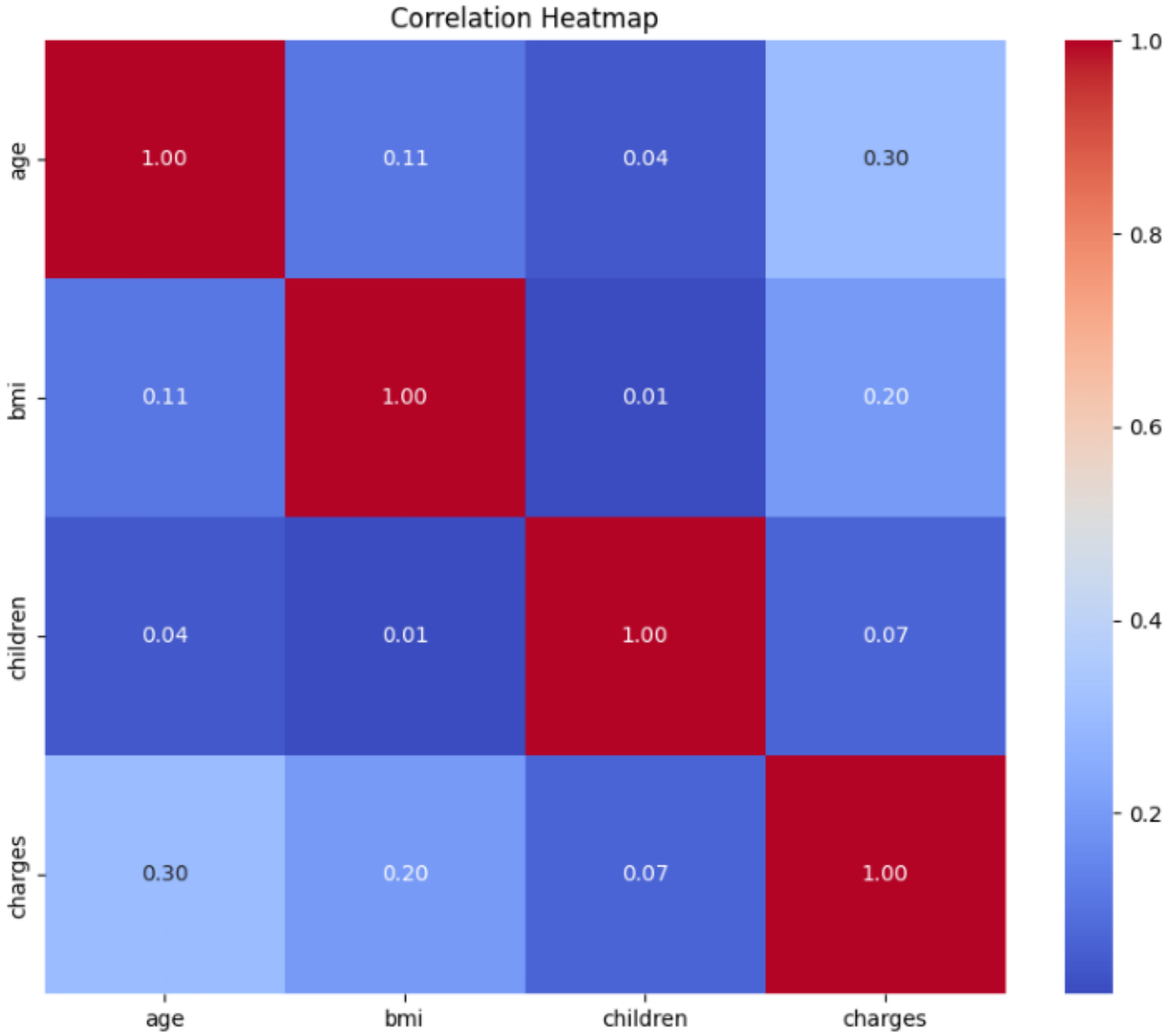


Figure 4. Correlation matrix of the numeric features.

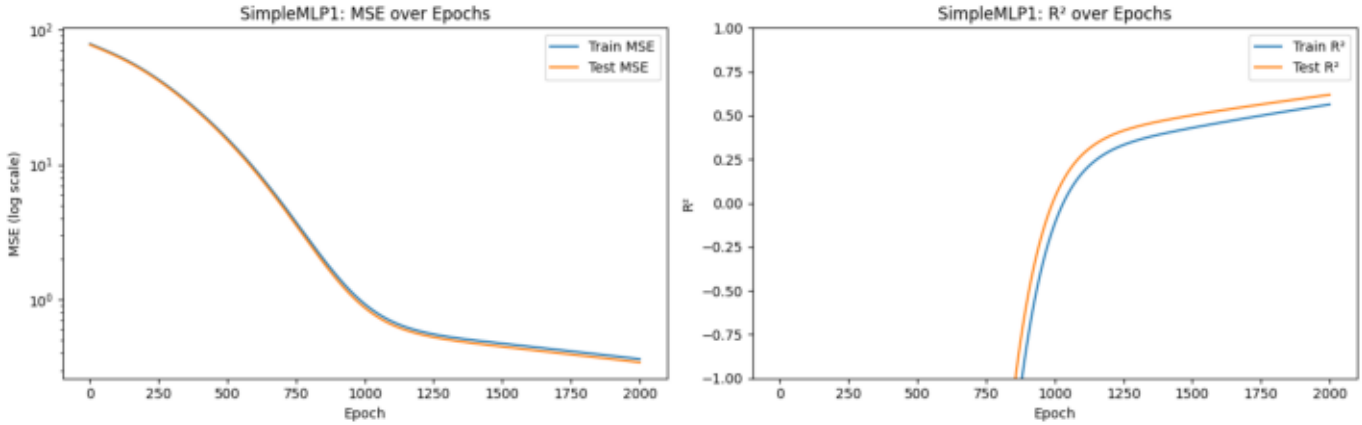


Figure 5. Training curves for SimpleMLP1 (log-charges target).

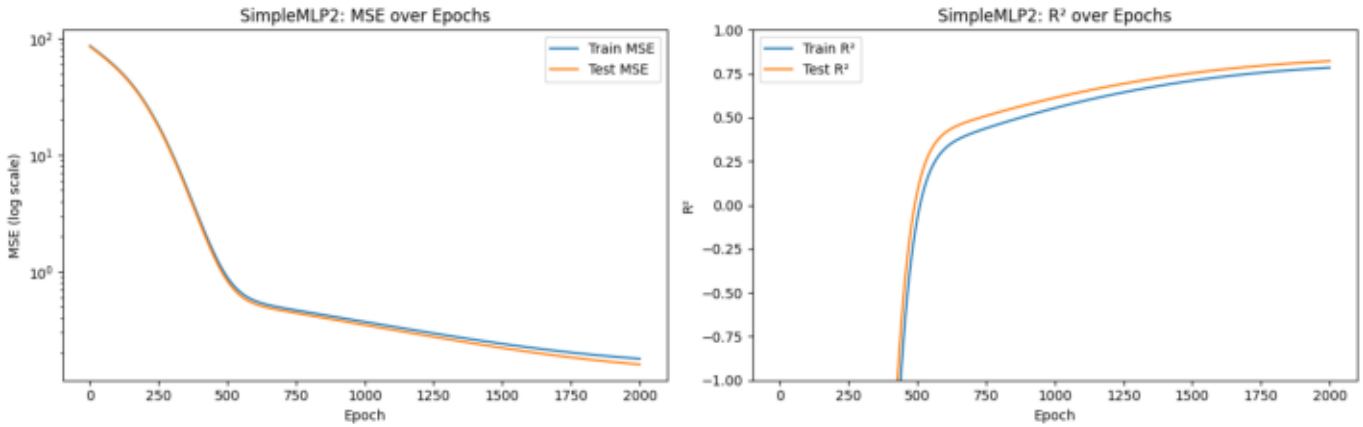


Figure 6. Training curves for SimpleMLP2 (log-charges target).

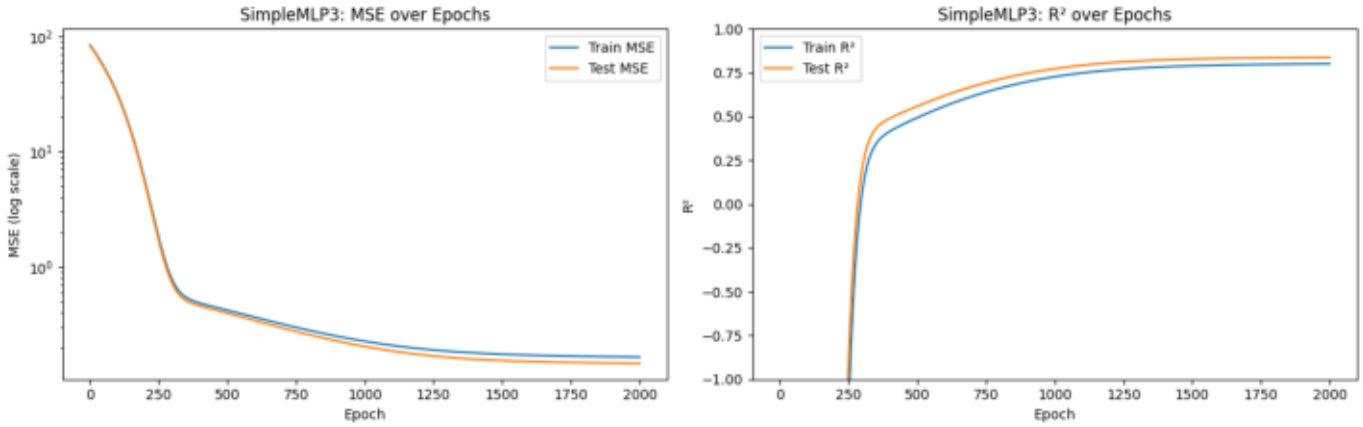


Figure 7. Training curves for SimpleMLP3 (log-charges target).

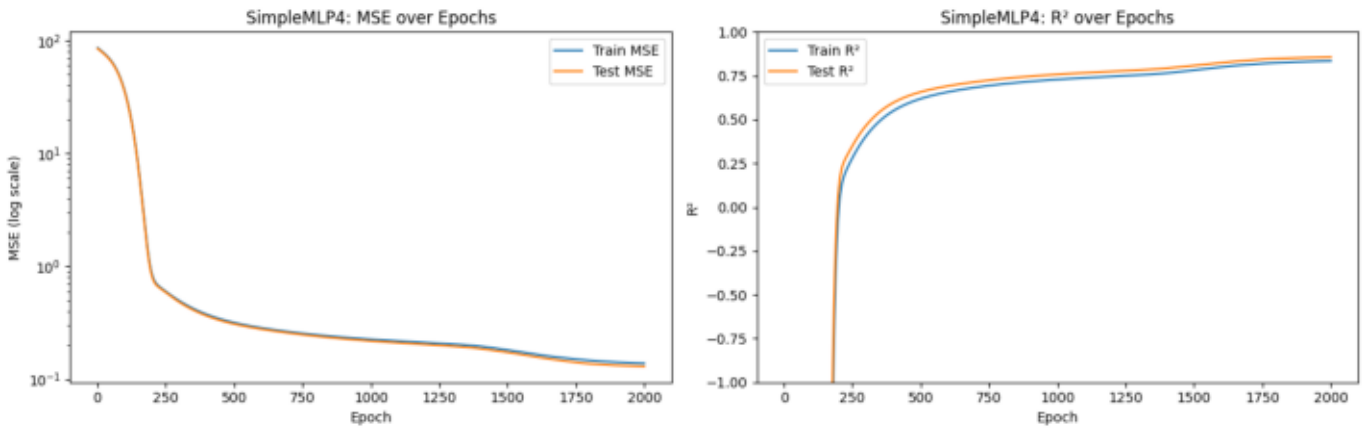


Figure 8. Training curves for SimpleMLP4 (log-charges target).

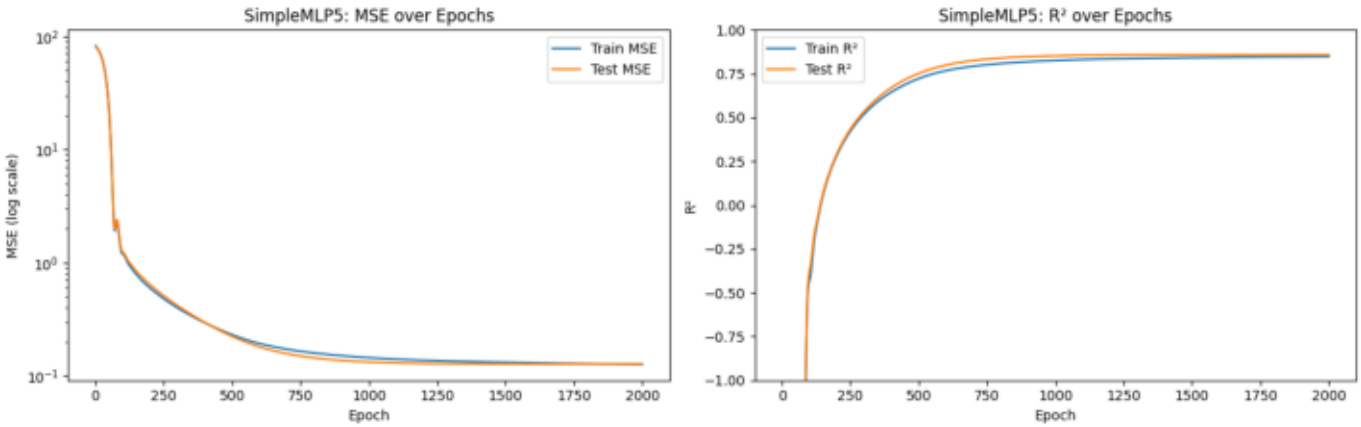


Figure 9. Training curves for SimpleMLP5 (log-charges target).

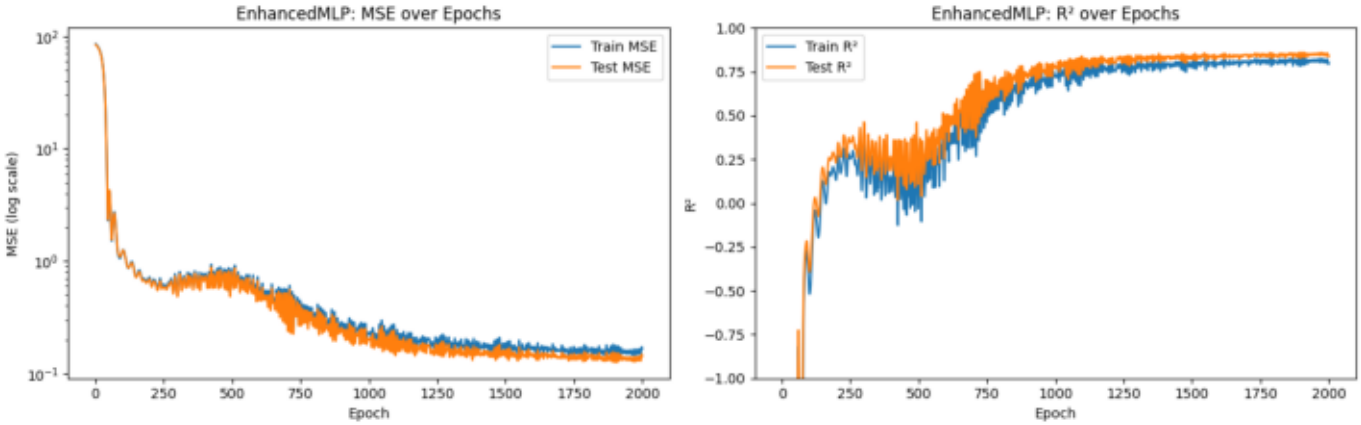


Figure 10. Training curves for EnhancedMLP (log-charges target).

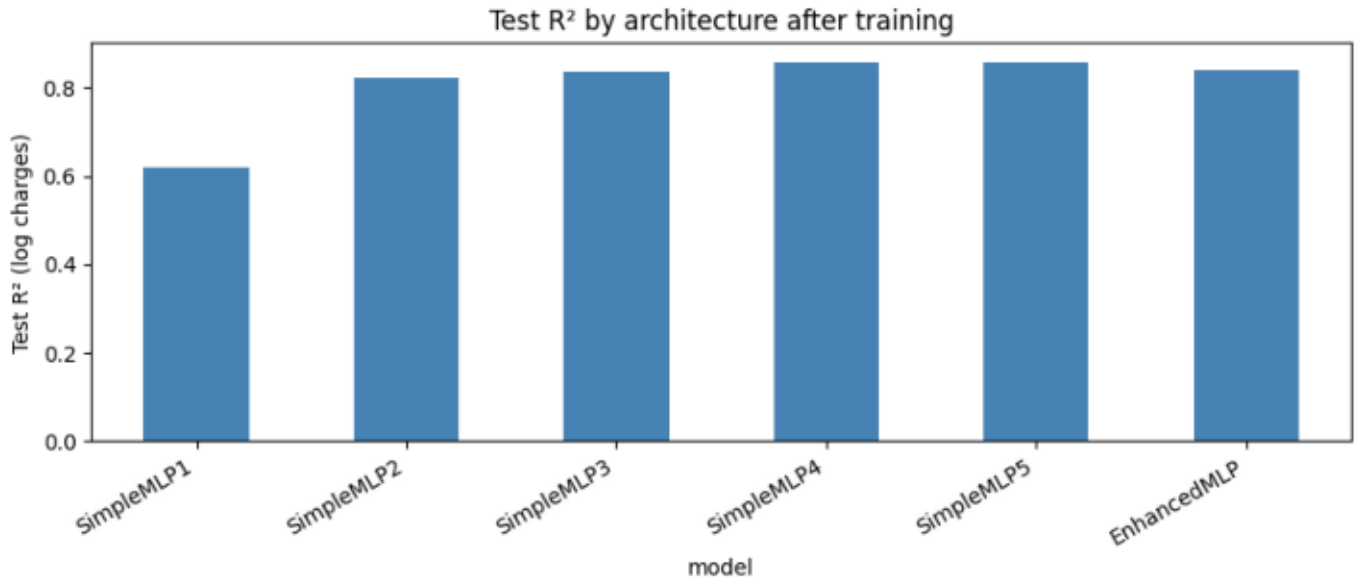


Figure 11. Test R² of each architecture after training.

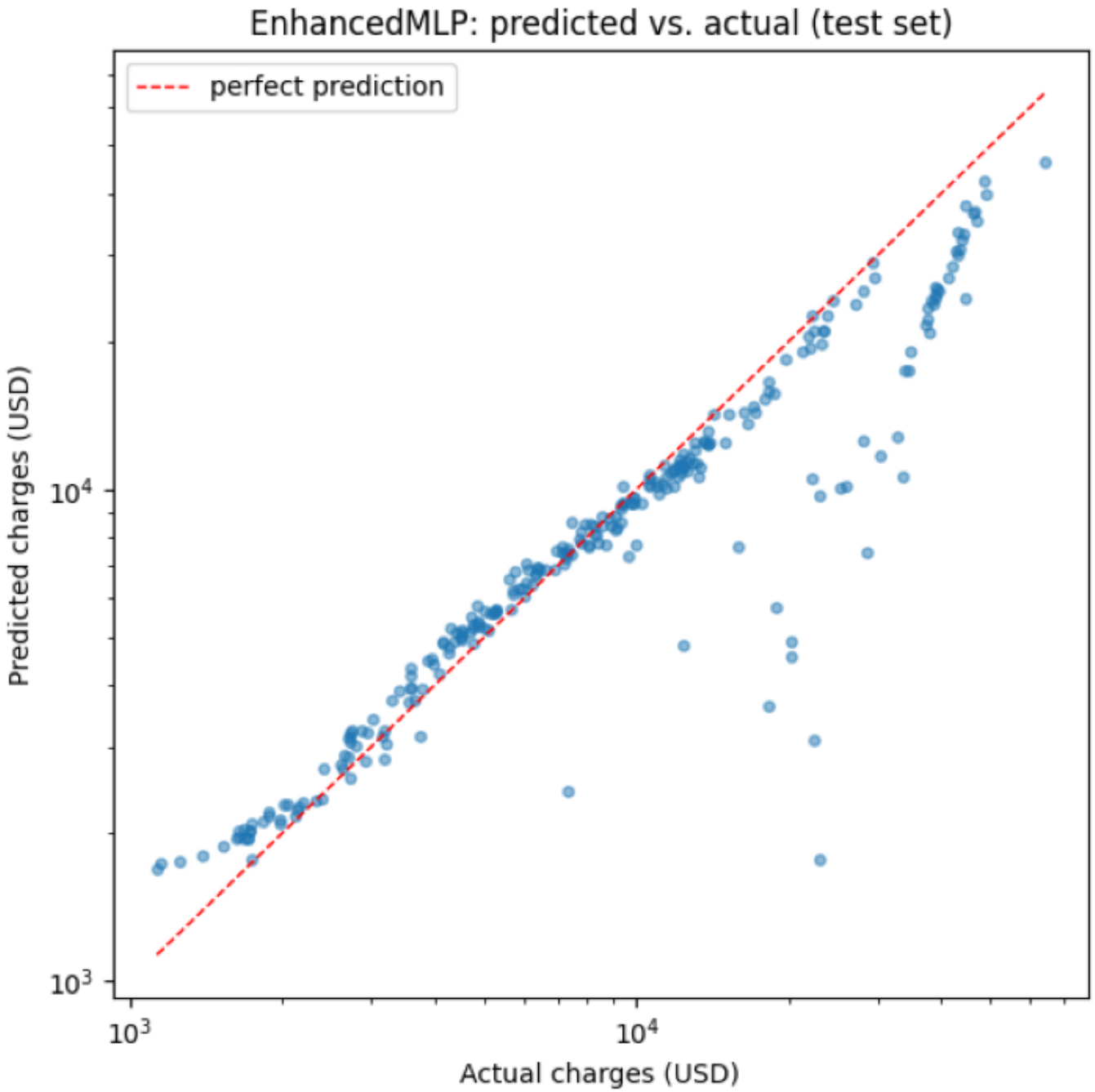


Figure 12. EnhancedMLP predictions vs. actual charges on the test set.